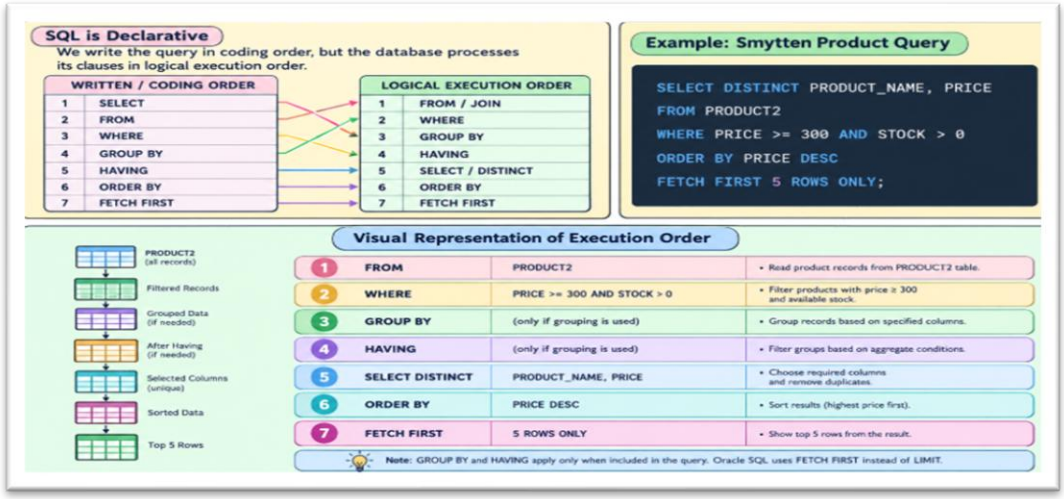
Smytten E-Commerce Database Management System

CHAPTER – 8

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products:

```
SQL> SELECT * FROM PRODUCT2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID  SELLER_ID
---------- ---------- ----------- ----------
       101
Face Wash
       349         50           1          1

       102
Moisturizer
       399         40           1          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID  SELLER_ID
---------- ---------- ----------- ----------

       103
Lipstick
       499         35           2          2

       104
Mascara

PRODUCT_ID
----------
```

Display selected product details:

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  WHERE PRICE < 500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Face Wash
       349

       102
Moisturizer
       399

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Lipstick
       499

       104
Mascara
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  WHERE PRICE BETWEEN 300 AND 600;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Face Wash
       349

       102
Moisturizer
       399

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Lipstick
       499

       104
Mascara
```

Find product id and name where category id is 1.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, CATEGORY_ID
  2  FROM PRODUCT2
  3  WHERE CATEGORY_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       101
Face Wash
       349           1

       102
Moisturizer
       399           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------

       109
Face Serum
       799           1
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  ORDER BY PRICE ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Face Wash
       349

       105
Shampoo
       349

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       110
Conditioner
       379

       102
Moisturizer
```

Descending order of price

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       109
Face Serum
       799

       106
Hair Serum
       599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Lipstick
       499

       104
Mascara
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

```
SQL> SELECT DISTINCT CATEGORY_ID
  2  FROM PRODUCT2;

CATEGORY_ID
-----------
          1
          2
          3
```

```
SQL> SELECT DISTINCT CATEGORY_NAME
  2  FROM CATEGORY2;

CATEGORY_NAME
--------------------------------------------------------------------------------
Fragrance
Haircare
Makeup
Skincare
Wellness

SQL> SELECT * FROM CUSTOMER2;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
--------------
ADDRESS
--------------------------------------------------------------------------------
       101
Haripriya
haripriya@gmail.com
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE_NUMBER
  2  FROM CUSTOMER2;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
         101
Haripriya
haripriya@gmail.com
9876543210


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
         102
Ananya
ananya@gmail.com
9876543211
```

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, CATEGORY_ID
  2  FROM PRODUCT2
  3  WHERE CATEGORY_ID IN (1, 2);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       101
Face Wash
       349           1

       102
Moisturizer
       399           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------

       103
Lipstick
       499           2

       104
Mascara

PRODUCT_ID
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT Customer_ID, Customer_Name, Email, Contact_No
  2  FROM Customer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
CONTACT_NO
---------------
        101
Anitha Kumar
anitha@gmail.com
9876501234

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
CONTACT_NO
---------------
        102
Rahul
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, CATEGORY_ID
  2  FROM PRODUCT2
  3  WHERE CATEGORY_ID IN (1, 2);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       101
Face Wash
       349           1

       102
Moisturizer
       399           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------

       103
Lipstick
       499           2

       104
Mascara

PRODUCT_ID
```

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  WHERE PRICE < 300
  4  OR PRICE > 600;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       109
Face Serum
       799
```

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Organic Turmeric Powder
       130

       102
Black Pepper
       180

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  O.TOTAL_AMOUNT,
  6  O.ORDER_STATUS
  7  FROM CUSTOMER2 C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  ORDER BY O.ORDER_DATE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ ------------------------------
        101
Haripriya
       211 23-SEP-26          899 Shipped

        110
Aishwarya
       210 18-SEP-26          749 Pending
```

5.2 Customer Order Report

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT2
  3  WHERE PRICE >= 500
  4  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       109
Face Serum
       799

       106
Hair Serum
       599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

5.3 Product Order Report

```
SQL> SELECT
  2       od.Order_ID,
  3       od.Product_ID,
  4       od.Quantity,
  5       od.Unit_Price,
  6       od.Subtotal
  7  FROM Order_Details od
  8  ORDER BY od.Subtotal DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
       203        104          2       1600       3200
       201        101          3       1000       3000
       202        103          2        900       1800
       204        101          1       1500       1500
       201        102          1        500        500
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.Customer_Name,
  4       COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  JOIN Orders o
  7       ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY Total_Orders DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
       103
Priya
           1

       102
Rahul
           1

CUSTOMER_ID
-----------
CUSTOMER_NAME
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.Customer_Name,
  4       SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7       ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
       103
Priya
          3200

       101
Anitha Kumar
          2700

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
```